Hugo Marquerie 15/06/2025

Teorema de convergencia de martingalas

Teorema 1 (de convergencia de martingalas). *Sea $(X_n)_n$ una submartingala*

$$\left(\sup_{n\in\mathbb{N}} \|X_n\|_1 < \infty\right) \vee \left(\forall n \in \mathbb{N} : X_n \leqslant 0\right) \implies X_n \xrightarrow[n\to\infty]{c.s.} X_\infty \wedge |X_\infty| < \infty \ c.s.$$

***Demostración*:** Vamos a dividir la prueba en casos:

I. Consideramos $\forall n \in \mathbb{N} : X_n \geqslant 0$ y $\sup_{n\in\mathbb{N}} \|X_n\|_2 < \infty$.

(a) Tenemos que para todo $k, n \in \mathbb{N}$,

$$\begin{aligned}\|X_{n+k} - X_n\|_2^2 &= \mathbb{E}\left[X_{n+k}^2\right] + \mathbb{E}\left[X_n^2\right] - 2\mathbb{E}\left[X_{n+k}X_n\right] \\ &= \mathbb{E}\left[X_{n+k}^2\right] + \mathbb{E}\left[X_n^2\right] - 2\mathbb{E}\left[\mathbb{E}\left[X_{n+k}X_n \mid \mathcal{F}_n\right]\right] \\ &= \mathbb{E}\left[X_{n+k}^2\right] + \mathbb{E}\left[X_n^2\right] - 2\mathbb{E}\left[X_n \cdot \mathbb{E}\left[X_{n+k} \mid \mathcal{F}_n\right]\right]\end{aligned}$$

Ahora, $\mathbb{E}\left[X_{n+k} \mid \mathcal{F}_n\right] = \mathbb{E}\left[\mathbb{E}\left[X_{n+k} \mid \mathcal{F}_{n+k-1}\right] \mid \mathcal{F}_n\right] \geqslant \mathbb{E}\left[X_{n+k-1} \mid \mathcal{F}_n\right]$. Iterando k veces, tenemos que $\mathbb{E}\left[X_j \mid \mathcal{F}_k\right] \geqslant X_k$ para todo $j > k$. Por tanto,

$$\|X_{n+k} - X_n\|_2^2 \leqslant \mathbb{E}\left[X_{n+k}^2\right] + \mathbb{E}\left[X_n^2\right] - 2\mathbb{E}\left[X_n^2\right] = \mathbb{E}\left[X_{n+k}^2\right] - \mathbb{E}\left[X_n^2\right].$$

Entonces, deducimos que

i. La sucesión $\left(\|X_n\|_2\right)_{n\in\mathbb{N}}$ es creciente.

ii. $\|X_n\|_2 \nearrow \sup_{n\in\mathbb{N}} \|X_n\|_2$.

iii. $(X_n)_{n\in\mathbb{N}}$ es una sucesión de Cauchy en $\mathcal{L}^2(\mathbb{P})$. Como $\mathcal{L}^2(\mathbb{P})$ es completo,

$$\exists X_\infty \in \mathcal{L}^2(\mathbb{P}) : X_n \xrightarrow[n\to\infty]{\mathcal{L}^2} X_\infty \text{ con } X_\infty < \infty \text{ c.s.}$$

(b) Sea $(n_k)_{k\in\mathbb{N}}$ tal que $\|X_{n_k} - X_\infty\|_2 \leqslant 2^{-k}$. Sea $\varepsilon > 0$, definimos $A_k^\varepsilon = \{|X_{n_k} - X_\infty| > \varepsilon\}$.

$$\begin{aligned}\implies \sum_{k=1}^\infty \mathbb{P}\left(A_k^\varepsilon\right) &\overset{\substack{\text{Chebyshev}\\ \downarrow}}{\leqslant} \frac{1}{\varepsilon^2}\sum_{k=1}^\infty \mathbb{E}\left[|X_{n_k} - X_\infty|^2\right] = \\ &\leqslant \frac{1}{\varepsilon^2}\sum_{k=1}^\infty \|X_{n_k} - X_\infty\|_2^2 \leqslant \frac{1}{\varepsilon^2}\sum_{k=1}^\infty 4^{-k} = \frac{4}{3\varepsilon^2} < \infty.\end{aligned}$$

Entonces, por el lema de Borel-Cantelli I, $\mathbb{P}\left(\limsup_{k\to\infty} A_k^{\varepsilon}\right) = 0 \implies X_{n_k} \xrightarrow[k\to\infty]{\text{c.s.}} X_\infty$.

(c) Tenemos que

$$\begin{aligned}\left\|X_{n_{k+1}} - X_{n_k}\right\|_1 &\leqslant \left\|X_{n_{k+1}} - X_\infty\right\|_1 + \left\|X_\infty - X_{n_k}\right\|_1 \\ &\leqslant \left\|X_{n_{k+1}} - X_\infty\right\|_2 + \left\|X_\infty - X_{n_k}\right\|_2 \\ &\leqslant 2^{-k-1} + 2^{-k} = 3 \cdot 2^{-k-1}\end{aligned}$$

Para cada k fijo, definimos, para $j \geqslant 0$, $Y_j = X_{n_k+j} - X_{n_k}$ que es una submartingala con respecto a la filtración $\left(\mathcal{F}_{n_k+j}\right)_{j\in\mathbb{N}}$. Entonces, por el corolario,

$$\mathbb{P}\Bigg(\underbrace{\left\{\omega \in \Omega : \max_{n_k \leqslant j < n_{k+1}} |Y_j| > \varepsilon\right\}}_{B_k^{\varepsilon}}\Bigg) \leqslant \frac{2}{\varepsilon} \cdot \mathbb{E}\left[|Y_{n_k+1}|\right].$$

Sumando en k, tenemos que

$$\begin{aligned}\sum_{k=1}^{\infty} \mathbb{P}\left(B_k^{\varepsilon}\right) &\leqslant \frac{2}{\varepsilon} \sum_{k=1}^{\infty} \mathbb{E}\left[\left|Y_{n_{k+1}-n_k}\right|\right] = \\ &\leqslant \frac{2}{\varepsilon} \sum_{k=1}^{\infty} \left\|X_{n_k+1} - X_{n_k}\right\|_1 \leqslant \frac{6}{\varepsilon} \sum_{k=1}^{\infty} 2^{-k-1} = \frac{3}{2\varepsilon} < \infty.\end{aligned}$$

De nuevo por el lema de Borel-Cantelli I, $\mathbb{P}\left(\limsup_{k\to\infty} B_k^{\varepsilon}\right) = 0$. Entonces, si definimos $Z_k = \max_{n_k \leqslant j < n_{k+1}} |Y_j|$, tenemos que $Z_k \xrightarrow[k\to\infty]{\text{c.s.}} 0$.

(d) Sea $\omega \in \Omega$ tal que $X_{n_k}(\omega) \xrightarrow[k\to\infty]{} X_\infty(\omega)$ y $Z_k(\omega) \xrightarrow[k\to\infty]{} 0$.

Para cada $j \in \mathbb{N}$, podemos elegir $n_k(j)$ tal que $n_k(j) \leqslant j < n_{k+1}$

$$\implies |X_j(\omega) - X_\infty(\omega)| \leqslant$$

Como $Z_k \xrightarrow[k\to\infty]{\text{c.s.}} 0$ y $X_{n_k} \xrightarrow[k\to\infty]{\text{c.s.}} X_\infty$, entonces

$$\mathbb{P}\left(\{\omega : Z_k\}\right)$$

II. Consideramos $\forall n \in \mathbb{N} : X_n \leqslant 0$ c.s. Sea $Y_n = e^{X_n} \geqslant 0$.

$$\implies \left\|Y_n\right\|_\infty \leqslant 1 \implies \left\|Y_n\right\|_2 \leqslant 1.$$

$\blacksquare$

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.